Assignment 3 — RAG Application PRD

IIT Madras Student Knowledge Assistant

A citation-first AI assistant for answering student questions from official IIT Madras sources.

Product	IITM Knowledge Assistant
Primary user	Current IIT Madras undergraduate students
Knowledge base	Official IIT Madras student, academic, hostel and policy sources
Interface	Responsive web chat with citations
MVP scope	Information retrieval and explanation; no transactions or private records
RAG approach	Hybrid retrieval → reranking → grounded generation

Executive summary. IIT Madras students often need to locate a rule, deadline, fee instruction or hostel policy across multiple official pages, PDFs and service portals. The product retrieves the relevant passages, answers in plain language, and shows exactly where the answer came from. The official IIT Madras student hub itself points users to separate resources for courses, timetables, calendars, curricula, fees, hostel fees and portals. ■cite■turn1search1■

Product thesis. This is an information-retrieval product first and a chatbot second. RAG is appropriate because the corpus changes, source provenance matters, and the system should abstain rather than invent information.

Submission note. The assignment requires an actual screenshot from the Section 7 builder test. This draft includes the complete Build Brief but intentionally does not claim a mockup is an actual Lovable/Claude run.

1. Problem & User

One-line description: Ask IIT Madras student questions and get cited answers from official institute sources.

Persona: Current IIT Madras undergraduate students who need academic or administrative information at least once a week and currently search multiple IITM websites, PDFs and portals.

Today's workflow

- **Formulate:** convert a natural-language question into keywords. *Slow.*
- **Search:** use Google or IITM pages and open several results. *Slow.*
- **Interpret:** read a policy/PDF and identify the applicable clause. *Error-prone.*
- **Cross-check:** verify dates, exceptions or newer instructions on another page. *Error-prone.*
- **Act:** follow a portal/link or contact an office if still uncertain. *Frustrating.*

Why RAG vs. prompting vs. fine-tuning

- **Prompting alone:** cannot reliably inject the current IITM corpus at answer time.
- **RAG:** retrieves current source passages and exposes citations; it fits changing policy/administrative knowledge.
- **Fine-tuning:** useful for behavior/style, but not a substitute for a frequently refreshed factual corpus.

MVP boundary

Public official IIT Madras information only. The assistant does not expose private student records, grades, fees owed, medical/disciplinary information or credentials, and it does not perform transactions.

The assignment asks for a specific persona, current workflow, and an explicit argument for RAG over simpler or more complex alternatives. ■filecite■turn0file0■L107-L112■

2. Use Cases & Scope

User stories

Priority	User story	Rationale
P0	As a student, I want to ask an academic-policy question so I can understand the rule without searching several documents.	Core RAG value.
P0	As a student, I want citations for every factual answer so I can verify it.	Trust/auditability.
P0	As a student, I want the assistant to say it does not know when sources do not support an answer.	Hallucination control.
P1	As a student, I want follow-up questions without repeating context.	Conversation usability.
P1	As a student, I want a direct official link to complete the next step.	Answer → action.
P2	As a student, I want to rate an answer so poor retrieval can be improved.	Evaluation data.

Primary use case A — academic/administrative lookup

Happy path: “Can I drop a registered course?” → retrieve the current rule → answer in 2–5 bullets → cite document/section → link to official source.

Edge case: two sources contain different dates. The assistant surfaces the conflict, prefers the newest effective official source where determinable, and directs the user to the responsible office rather than silently choosing.

Primary use case B — hostel/student services

Happy path: “Where can I find hostel mess rebate rules?” → retrieve the official CCW source → explain the rule → provide the link. CCW publishes hostel rules, student online services and mess-rebate resources. ■cite■turn1search7■

Edge case: a user asks for another student's hostel/account information → refuse and do not retrieve private records.

Refusals and handoffs

- Refuse requests for another person's personal data, grades, fees, credentials, medical information or disciplinary information.
- Hand off questions requiring an institute decision, exception, complaint resolution, transaction, or authenticated portal access.

3. Architecture

The assignment's required pipeline is Sources → Ingest → Parse/OCR → Chunk → Embed → Vector DB → Retrieve → Re-rank → LLM → Response, with logging/observability across the flow. ■filecite■turn0file0■L27-L35■

Architecture

OFFICIAL SOURCES	→	INGEST/PARSE	→	CHUNK + METADATA	→	EMBED	→	VECTOR DB
IITM pages + PDFs		HTML/PDF/OCR		400–800 tokens; heading-aware		text-embedding-3-small		pgvector

QUERY	→	HYBRID RETRIEVE	→	RERANK	→	GROUNDED LLM	→	ANSWER + CITATIONS
Student question		keyword + vector		top 20 → top 5		GPT-5.4 mini		source + section + link

Data sources and refresh

- IIT Madras student hub: courses, timetable, calendar, curricula, fees, hostel fees and service links. ■cite■turn1search1■
- Office of Dean (Students): student policies, hostel rules, ordinances/regulations and governance material. ■cite■turn1search0■
- Council of Wardens/Hostel Management: hostel rules, student services and mess-rebate resources. ■cite■turn1search7■
- Refresh public sources daily; detect changes by URL + content hash; re-embed only changed documents.

Chunking: 400–800 tokens with 10–15% overlap, preserving heading hierarchy and metadata such as source title, URL, section, effective/update date and access scope.

Embedding: **text-embedding-3-small**; current published price is $0.02 per 1M input tokens and the model is designed for semantic search. ■cite■turn2search0■turn2search1■

LLM: **GPT-5.4 mini**; current published API price is $0.75/1M input and $4.50/1M output. ■cite■turn0search5■turn0search7■

Observability: request ID, anonymized session ID, query, retrieved chunk IDs/scores, reranker scores, token counts, model version, latency, citation IDs and feedback. Never log passwords or private student records.

4. Failure Modes & Fixes

Failure	Breaks	Detection	Fix
Bad source data	Outdated, duplicated, contradictory or mangled policy.	Extraction errors, hash/duplicate checks, human review of priority docs.	Prefer canonical source; clean headers/footers; retain version/effective-date metadata.
Wrong chunking	Clause split from its exception or heading.	Recall@5 drops; citation is incomplete.	Heading-aware chunks; overlap; inspect high-risk policy documents.
Retrieval miss	Correct passage not in top-K.	Recall@5/20 on golden set.	Hybrid retrieval, query expansion, larger candidate pool, reranking.
LLM hallucination	Model adds unsupported conditions.	Faithfulness score; citation mismatch.	Strict prompt; source-only answer; mandatory citation; abstention.
Knowledge stale	Index lags changed IITM source.	Freshness SLA; content-hash mismatch; ingestion alerts.	Daily refresh; re-index priority sources immediately; display source date.
User misuse	Ambiguous query, PII request, or low trust.	Query classifier; refusal rate; feedback reasons.	Clarifying question; privacy refusal; examples; citations and office handoff.

Highest-risk failure: a confident answer based on an outdated or incorrect policy. Therefore source freshness, citation correctness and abstention are core product quality metrics—not cleanup items.

The assignment explicitly expects each of the six failures to be product-specific and to include what breaks, how it is detected, and how it is fixed. ■filecite■turn0file0■L147-L183■

5. Evaluation, Permissions & Cost

Evaluation

- **Recall@5 ≥ 90%:** correct supporting passage appears in top five.
- **Faithfulness ≥ 95%:** answer claims are supported by cited sources.
- **Citation correctness ≥ 95%:** cited source actually supports the claim.
- **Useful-answer rate ≥ 85%:** pilot users mark answers useful without a second search.

Golden evaluation set — 8 questions

#	Question	Test
1	Can I drop a registered course?	Current academic rule.
2	Where is the academic calendar?	Official calendar link.
3	How do I pay hostel fees?	Official payment resource.
4	What are mess rebate rules?	Current CCW rule.
5	Which office handles hostels?	Official office evidence.
6	What is another student's CGPA?	Privacy refusal.
7	Can I get an exception to a rule?	Escalation.
8	fees	Ambiguity/clarification.

Who grades: automated retrieval metrics for every release plus human review. LLM-as-a-Judge can scale faithfulness checks, but 10–20% of judge-positive results should be human-audited.

Permissions

MVP uses public sources only, so there is no per-user document authorization. If authenticated sources are added later, chunks must carry access-control metadata and retrieval must filter by authorization before generation. The LLM is never the authorization layer.

Cost model

Assumption: 10,000 queries/month; 1,500 input tokens + 250 output tokens/query; one query embedding/query. At current GPT-5.4 mini pricing, 15M input tokens cost $11.25 and 2.5M output tokens cost $11.25. Query embeddings cost 15M × $0.02/M = $0.30. **Model/API subtotal ≈ $22.80/month**, excluding vector DB, hosting, monitoring and network. ■cite■turn0search5■turn2search0■

One-time ingestion of 5M tokens costs about $0.10 for embeddings at the same rate. The key optimization lever is caching repeated high-frequency answers and routing simple queries to a cheaper model/classifier before generation.

6. UX & Rollout

Interface

Responsive web chat because students already use the web to reach IITM information. The UI contains one prominent question box, suggested questions, concise answer, expandable citations, and a “Contact the relevant office” action when the assistant cannot answer.

Answer format

- Direct answer in 2–5 bullets.
- Source block with document/page/section and official link.
- Visible uncertainty message when evidence is incomplete or conflicting.
- Optional follow-up suggestions.

Low confidence

If no source passes the grounding threshold: “I couldn't find an authoritative IIT Madras source for that. I don't want to guess.” Then provide a relevant office/link if available. Never fabricate a citation.

Feedback

- Thumbs up/down on every answer.
- Down-vote reason tags: wrong source, incomplete, outdated, unclear, should have refused.
- Feed recurring failures into the golden test set.

Launch plan

Phase	Scope	Go / no-go
Internal alpha	50–100 golden questions + 5 student testers.	Recall@5 ≥90%; citation correctness ≥90%.
Closed beta	100–300 students; public corpus only.	Useful-answer rate ≥85%; zero critical privacy failures.
GA	Broader audience; daily ingestion monitoring.	Citation correctness ≥95%; ≥99% scheduled refresh success.

The first release does **not** integrate with the student portal or perform transactions. Transactional features are a later phase only after retrieval, citation and permission behavior are stable.

7. AI Build Brief

Platform: Lovable. Use a prepared document set of public IITM sources for the first prototype.

System prompt

You are the IITM Knowledge Assistant. Answer student questions using ONLY the supplied IIT Madras knowledge base. Retrieve and explain authoritative institute information clearly and briefly. Every factual answer MUST include citations naming the source document/page or section and a link when available. Never invent a rule, deadline, fee, office decision, eligibility condition or exception. If the retrieved sources do not support an answer, say: “I couldn't find an authoritative IIT Madras source for that, so I don't want to guess.” If sources conflict, state that they conflict and prefer the newest effective official source when determinable. Ask a clarifying question when the query is too ambiguous. Refuse requests for another person's personal data, grades, fees, credentials, medical information or disciplinary information. Do not perform transactions or claim an action is complete. For requests requiring an institute decision or exception, explain the published rule and direct the user to the responsible office. Keep normal answers under 150 words unless asked for more detail. Use plain, professional language and preserve official terminology where it matters.

Knowledge base

- Official IITM student resource hub: courses, timetable, calendar, curricula, fees, hostel fees and services. ■cite■turn1search1■
- Office of Dean (Students): policies, hostel rules, ordinances/regulations. ■cite■turn1search0■
- CCW/Hostel Management: rules, student services, mess rebate resources. ■cite■turn1search7■

MUST / SHOULD / COULD

MUST	SHOULD	COULD
Retrieve before answering	Show source snippets	Related-question suggestions
Cite every factual answer	Show source freshness/date	Multilingual UI
Refuse unsupported/private requests	Collect structured feedback	Voice input
Hybrid retrieval + reranking	Show uncertainty state	Authenticated features later
Trace retrieval/citations without PII	Provide direct source links	Personalized student context later

Example interactions

Happy path: User: “Where can I find hostel mess rebate rules?” Assistant: give the relevant rule in concise language and cite the official CCW source.

Refusal: User: “What is Rahul's CGPA?” Assistant: refuse to provide another student's private academic information.

Escalation: User: “I missed the deadline. Can IIT Madras make an exception?” Assistant: explain the published rule if available, but state that only the responsible institute office can decide an exception.

Build validation: Run this brief in Lovable/Claude and capture the actual screenshot required by the assignment. Add 2–3 sentences on what matched the specification and what did not.

Source Notes & Submission Checklist

Assignment basis

The uploaded brief requires one 6–10 page PDF with seven sections, an architecture diagram, real cost math, a complete system prompt and an actual Build Brief test screenshot. ■filecite■turn0file0■L53-L95■ ■filecite■turn0file0■L281-L303■

External sources

- IIT Madras — Students: official academic/service resource hub. ■cite■turn1search1■
- IIT Madras — Office of Dean (Students): student policies and regulations. ■cite■turn1search0■
- IIT Madras — Council of Wardens & Hostel Management: hostel rules/services. ■cite■turn1search7■
- OpenAI — GPT-5.4 mini model/pricing. ■cite■turn0search5■turn0search7■
- OpenAI — text-embedding-3-small pricing and embeddings guide. ■cite■turn2search0■turn2search1■

Final submission checklist

- 6–10 page PDF.
- Architecture diagram.
- Six product-specific failure modes with detection and fixes.
- Evaluation metrics and golden set.
- Permission model.
- Current pricing and cost math.
- Complete system prompt, MUST/SHOULD/COULD and three example interactions.
- Actual builder screenshot + 2–3 sentence test reflection.

> The product deliberately keeps V1 narrow: answer questions from authoritative public IIT Madras sources, cite them, and abstain when evidence is insufficient. This makes the RAG behavior measurable before introducing transactional or personalized features.